

The Data Monetisation team builds privacy-safe customer signals and audience segments that enable partners to generate anonymised insights, improve AI models, and run compliant marketing campaigns.

You are a **Data Engineer** on this team. Your task is to design a new strategic audience segment called **Early_Tech_Adopters** —customers who buy or show strong interest in newly launched technology shortly after release.

Complete the following tasks by referring the table schemas provided below.

Task 1 — Define the segment rules and produce the segment

Write a clear, auditable SQL query that returns **one row per user** with the following fields:

- `user_id`
- `early_adopter_flag` (TRUE/FALSE)
- `evidence_source` (which rule(s) triggered the flag)

A user should be flagged as an early adopter if one or more of the following signals apply:

1. **Purchase behaviour:** Purchased a “latest” SKU within *90* days of launch.
2. **Interest behaviour:** Viewed the same “latest” SKU at least **2 times within 30 days** of launch.
3. **Profile support (secondary evidence):** Signals such as `prefers_new_releases = TRUE` or `trade_in_member = TRUE`.

Task 2 — Sanity-check the segment

Produce a quick validation query to sanity-check your output. This should include, for example:

- **Count of early adopters and share of total active users** (users with any event in the last 90 days).
- A **breakdown by evidence_source** and **product category** (e.g., smartphone vs laptop).

Task 3 — Explain the logic in non-technical terms

Provide **2–3 lines of plain-English explanation** describing:

- How your rules identify “early tech” behaviour
- Why the logic is sensible for commercial monetisation
- How recency, repeated interest, and purchase evidence support the definition.

Mock Schema

```
-- Product catalog attributes
-- launch_date: official launch date of the SKU
-- category: 'smartphone' | 'laptop' | 'tablet' | 'wearable' | 'accessory' ...
CREATE TABLE product_catalog (
  sku STRING,
  product_name STRING,
  category STRING,
  launch_date DATE
);

-- SKU pageviews
CREATE TABLE sku_views (
  user_id STRING,
  event_ts TIMESTAMP,
  sku STRING
);

-- Orders (web or store), one row per purchased SKU
CREATE TABLE orders (
  user_id STRING,
  order_ts TIMESTAMP,
  sku STRING,
  price NUMERIC
);

-- Optional profile signals (self-declared)
CREATE TABLE profiles (
  user_id STRING,
  snapshot_date DATE,
  prefers_new_releases BOOL, -- user opted into 'new releases' comms
  trade_in_member BOOL      -- enrolled in device trade-in programme
);
```

Sample Data

```
-- product_catalog (sample)
sku      | product_name | category  | launch_date
-----+-----+-----+-----
P001     | Phone X 2026 | smartphone | 2026-02-01
P002     | Phone X 2025 | smartphone | 2025-07-01
L010     | UltraBook 16 | laptop    | 2026-01-15

-- sku_views (sample)
user_id | event_ts | sku
```

U1		2026-02-02 09:02:00 UTC		P001
U1		2026-02-02 09:10:00 UTC		L010
U2		2026-02-25 18:22:00 UTC		P002

-- orders (sample)

user_id		order_ts		sku		price
U1		2026-02-05 11:30:00 UTC		P001		999
U3		2026-02-20 15:40:00 UTC		P002		699

-- profiles (sample)

user_id		snapshot_date		prefers_new_releases		trade_in_member
U1		2026-02-10		TRUE		TRUE
U2		2026-02-10		NULL		FALSE
U3		2026-02-10		FALSE		TRUE
